(U) ModelNameGoesHere
Architecture Description Document



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(U) FOR TITLE PAGE
(U) TIMELINE: MOST ITEMS: Ready by SRR, finalized by PDR. REVISION NUMBERS & DATES: Finalized at FDR.
(U) "Prepared By" and "Classified By" must be a company name, not an individual's name
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(U) FOR REVISION TABLE:
(U) TIMELINE: Each milestone submission needs its own row, in 0.# format. Final delivery is 1.0. Dates are YYYYMMDD.

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Revision	Date	Author	Description of Changes
1.0	201609	NGIC (Everett)	Initial Draft
2.0	202304	NGIC (Everett)	Revised for new Projects
2.1	20240908	BAH	Revised for new projects and formatting
2.2	20241014	BAH	Revised based on feedback
2.3	20250117	BAH	Revised based on feedback
3.0	20250217	BAH	Baselined to 3.0 release for document templates
4.0	20251114	BAH	Baselined to 4.0 release for document templates

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1 (U) Introduction

(U) TIMELINE: Should be finalized prior to SRR, and unless it's a brand new type of system, should be reasonably understood prior to kickoff. (U) This section provides a high-level overview of the document's purpose and scope. It defines the intended audiences—such as users, developers, and stakeholders—and explains how the document supports their understanding and use of the model. It establishes the foundation for all subsequent sections by outlining the intended use for the model and the structure of the documentation.

(U) The ModelNameGoesHere Architecture Description Document (ADD) discusses the codebase for the ModelNameGoesHere model developed by Booz Allen Hamilton (BAH) in the Overarching Dynamic Electronic-Warfare System Standard Architecture (ODESSA) framework in support of the National Ground Intelligence Center (NGIC). The information in this document outlines the computational decisions, frameworks, and development pathways that guided the development of this model. This document prepares users to understand the codebase at a much deeper computer science level, whereas the UAM is concerned more with user guidance, the underlying theoretical background and mathematics, and only broad-view discussion of the codebase.

1.1 (U) Threat System Description

The "Reference ..." sentence below directs the reader to the classified UAM instead of describing the system here.

(U) Reference the ModelNameGoesHere Users' and Analysts' Manual, Section #, for a description of the threat system.

1.2 (U) Architecture

1.2.1 (U) Architecture Selection Criteria

(U) TIMELINE: Should be finalized prior to System Requirements Review (SRR), and unless it's a brand new type of system, should be reasonably understood prior to kickoff.
(U) Based on the information presented at model kickoff and on the developer's research in the study phase, the developer should provide enough information in this section to support their proposal before SRR. Provide rationale for the specific Threat Modeling and Analysis Program (TMAP) architecture chosen (MATrix LABoratory (MATLAB)/Simulink, ODESSA, Command and Control (C2), etc.). From there additional details will be communicated on why specific version of ODESSA architecture, C2, or MATLAB/Simulink were chosen for this model. If an existing TMAP model was selected as a baseline, explain the selection rationale.

1.2.2 (U) Architecture Version

(U) TIMELINE: Should be finalized prior to SRR.
(U) Previous section focuses on the architecture chosen. This section focuses on which version of the architecture was chosen and why. A previous version may be chosen with justification and approval from NGIC Contracting Officer Representative (COR).

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1.2.3 (U) External Integration

(U) TIMELINE: Should be finalized prior to Preliminary Design Review (PDR).
(U) Please include any specific decisions that were made due to integration requirements here. Things like timing updates may have been chosen due to limitations of Integrated Threat Analysis and Simulation Environment (ITASE), include details on that here. This also may include requirements to interface with hardware, and it needs to be captured before the "Please reference the Interface Control Document (ICD)" comment below.
(U) This section should define the interfaces with other applications (ITASE, Trajectory Generator, Xpatch, database, file format, network connection, etc.), Product Components, and users as applicable. Include design constraints associated with interfaces. Ensure that the external interfaces are documented. If this is to be integrated into ITASE, include ITASE Application Programming Interface (API) and ICD discussion with any proposed modifications to the API. Include descriptions of architecture Graphical User Interface (GUI) uses and capabilities.
(U) If any third-party tools are *required* for the model to function or to create input data for the model, describe them here. If a third-party tool was used during development and/or testing to create input files, for output data processing, or for analysis, and the file format itself is the mandatory element and not that particular third party tool, explain that here.

(U) Please reference the ModelNameGoesHere ICD for more details on external interfaces and details on integration with other models or simulations.

1.3 (U) Standards and Regulations

1.3.1 (U) Architecture Standards and Regulations

(U) TIMELINE: Should be finalized prior to SRR.
(U) Identify any standards or regulations that may apply (e.g., Joint Technical Architecture Design, Airworthiness, ITASE Standards, Advanced Framework for Simulation, Integration, and Modeling (AFSIM) Standards, IEEE, etc.) beyond the architecture ADD. If none, just state there are none required.
(U) Leave text with reference to architecture document unless one does not exist.

Please reference the [architecture name here] ADD for details on standards and regulations for the architecture.

1.3.2 (U) Model Standards and Regulations

(U) TIMELINE: Should be finalized prior to SRR, and unless it's a brand new type of system, should be reasonably understood prior to kickoff. (U) This section identifies and outlines all relevant standards, regulatory frameworks, and compliance protocols applicable to the development, execution, and distribution of the model. It provides references to any governing military, industry, or organizational specifications which inform design constraints, performance benchmarks, interface definitions, and testing methodologies. Where applicable, this section highlights adherence to documentation standards, security classifications, data handling procedures, and model verification practices. The intent is to ensure the model aligns with externally imposed technical, procedural, and legal expectations, and to provide clarity for both users and developers on the guiding requirements that shape and constrain the model's design and deployment.

(U) This model and its associated documentation adhere to a defined set of technical, procedural, and security standards to ensure both consistency in development and compliance with governing requirements. The following

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subsections outline the guiding principles and regulations that have shaped this effort.

1.3.2.1 (U) Controlled Unclassified Information Compliance

(U) All model components and supporting documentation are developed in accordance with regulations governing Controlled Unclassified Information. In particular, the protection, storage, transmission, and dissemination of CUI material align with the requirements set forth by the National Archives and Records Administration (NARA) and Department of Defense (DoD) CUI implementation guidelines. Access and distribution controls are applied in accordance with marking, handling, and safeguarding procedures.

1.3.2.2 (U) NIST SP 800-171 Alignment

(U) As required for systems handling CUI within non-federal environments, this effort complies with the National Institute of Standards and Technology (NIST) Special Publication 800-171, "Protecting Controlled Unclassified Information in Nonfederal Systems and Organizations." Applicable security requirements are followed, including access control, audit and accountability, configuration management, identification and authentication, media protection, and system and information integrity.

1.3.2.3 (U) C++ Development Standards

(U) Code development for this model adheres strictly to the Google C++ Style Guide, ensuring consistency, readability, and maintainability throughout the codebase. All source files, class structures, function declarations, and variable naming conventions reflect this guidance, and developers are expected to follow formatting, exception handling, and memory management practices as outlined therein.

1.3.2.4 (U) Code Design Philosophy

(U) In addition to the structural guidance of the Google Style Guide, the model's implementation philosophy reflects the principles established by Bjarne Stroustrup, the creator of C++. These include strong emphasis on type safety, abstraction, modularity, and clarity of intent. Code is written to be maintainable and extensible, with careful consideration given to encapsulation, minimal coupling, and maximal cohesion, in line with modern C++ best practices.

(U) All developers contributing to this model are expected to be familiar with these standards and to apply them consistently across their contributions. Where deviations are necessary, justification and documentation must be provided to preserve long-term integrity and traceability within the system.

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1.3.3 (U) Supported Software and Operating Systems

(U) TIMELINE: Should be finalized prior to SRR, and unless it's a brand new type of system, should be reasonably understood prior to kickoff. (U) This section outlines the necessary prerequisites for executing the model effectively. It includes detailed specifications related to operating system compatibility, minimum and recommended hardware configurations, software dependencies, and any required input data formats. These requirements are essential for both proper installation and reliable performance of the model. Where applicable, guidance is provided on storage considerations, memory usage, processor requirements, and additional runtime environments (e.g., DLLs, drivers, or middleware). The intent is to ensure that users and developers can prepare and verify host systems appropriately prior to deployment.

(U) The ModelNameGoesHere model and corresponding external simulation Dynamic Link Library (DLL) will be compiled and tested in the following environment:

Table 1: (U) Minimum Computer Spec Suggestions

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Component	Specification
Processor	13th Gen Intel(R) Core(TM) i7-1370PX @ 1.90 GHz
Cores	14
Threads	20
RAM	32.0 GB (31.6 GB usable)
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2 (U) Model Architecture Description

(U) TIMELINE: Should be finalized prior to SRR. Note that this may update some over the development of the model, but they should be minimal. If new modes or operational changes are made, the NGIC model Technical Point of Contact (TPOC) should be notified.
(U) Provide here a written overview of the model architecture to summarize and supplement the Class Diagram.

2.1 (U) Threat Model Class Diagram

(U) Below is a Unified Modeling Language (UML) diagram, class diagram, or code flow diagram of the system model that defines how the various blocks of code connect to each other. This section should also contain descriptions that capture any specifics about the functional code which includes things like, "The system is broken out into two different entity controllers" or different system controllers, etc. Please describe any major attributes of the system code.
(U) The below example UML diagram is hard to read but included for completeness (shows the overall flow). Include zoom ins and additional images as needed (either below or in component and sub-component sections as appropriate) to demonstrate how the model is working.

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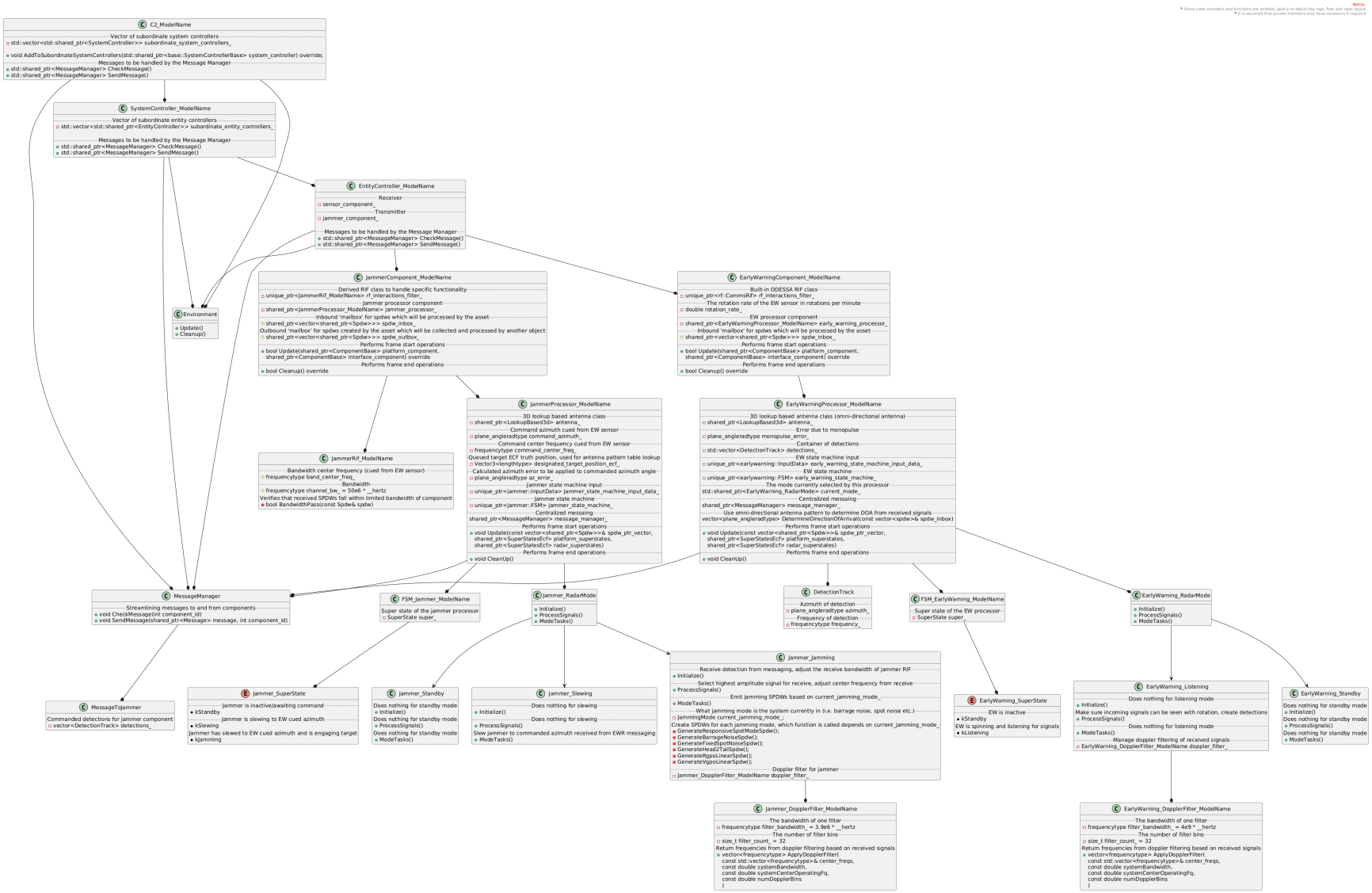


Figure 1: (U) ModelNameGoesHere UML

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2.2 (U) Sample Component Overview

(U) TIMELINE: Should be populated prior to PDR and finalized by Critical Design Review (CDR).
(U) In the context of this document, component refers to a functional block of code that is available within the software model that is representing a real-world capability or effect. In most cases, component refers to one of the blocks in the block diagram which is contained in the User and Analyst Manual (UAM) document and/or the code flow diagram contained within the ADD. Note for most cases, if there is a block in the code flow or block diagram, there should be a component (or multiple) section for each block.
(U) Describe the purpose of the Product Component and the rationale for developing, reusing, or refactoring the Product Component. For Product Components that will be reused or refactored, provide a reference to the existing design if applicable.

2.2.1 (U) Sample Component Code Structure

(U) TIMELINE: Should be populated by PDR and finalized by CDR.
(U) This section is focused on the code and use of the component within the model. Key information is which function(s) are associated with this component and data inputs and outputs that are coming from this component. Be careful not to conflate the component functionality description appropriate in the ADD with the system model details that are appropriate in the UAM.
(U) This section should describe all public interfaces to the component along with necessary pre-conditions of things external to the component and arguments passed to those interfaces for meaningful results. The behavior of the interface should be detailed enough to enable a developer utilizing those interfaces to understand the expected outcome of a proper use of the interface such as changes in state of the component, modifications to arguments passed by reference or pointer to the interface, or return values.
(U) Include algorithms (pseudocode), UAM sequence diagrams, statechart diagrams etc. as applicable to convey the information (i.e., it may not be necessary to document algorithms such as accessors, mutators, and other simple algorithms). If an equation is described in the UAM, the pseudocode should be described here. This section is focused on the computer science of the component details.
(U) Do NOT include code snippets as it can change over time and currently trying to keep overhead document costs down.
(U) EXAMPLE: This component can be instantiated multiple times to represent multiple antennas in the array, and the specific positioning of the individual antenna is specified in the input_param_struct." ... "One of the input parameters allows the developer to configure the antenna as a receiver or a transmitter."
(U) EXAMPLE: "This component returns a gain_struct relative to the antenna boresite, the pedastal component (described in section y.x) is responsible for providing the physical pointing direction, so the gain in a particular direction at any particular time in the simulation is calculated by taking the outputs from X and Y and interpolating..."
(U) EXAMPLE: If a component has a reference to the spdw_inbox that is not passed as part of an interface function and assumes that there are Simulated Pulse Descriptor Word (SPDW) there, this would be a precondition.
(U) EXAMPLE: If a list of items is passed by reference to an interface of the component which manipulates that list, that should be captured so the caller of the interface is aware. (U) EXAMPLE: This component cannot interact directly with an external sim, and so does not have a function described in the ICD. However, for deep dive analysis of the system model, the telemetry_struct can be logged.

3 (U) Sources

(U) Reference the UAM for a comprehensive list of sources used to develop this model. Reference Citations below for those sources used in creating this ADD.

4 (U) Acronyms

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Term	Description	Page
ADD	*Architecture Description Document*	1, 2, 6
AFSIM	*Advanced Framework for Simulation, Integration, and Modeling*	2
API	*Application Programming Interface*	2
BAH	*Booz Allen Hamilton*	1
C2	*Command and Control*	1
CDR	*Critical Design Review*	6
COR	*Contracting Officer Representative*	1
GUI	*Graphical User Interface*	2
ICD	*Interface Control Document*	2
ITASE	*Integrated Threat Analysis and Simulation Environment*	2
MATLAB	*MATrix LABoratory*	1
NGIC	*National Ground Intelligence Center*	1, 4
ODESSA	*Overarching Dynamic Electronic-Warfare System Standard Architecture*	1
PDR	*Preliminary Design Review*	2, 6
SPDW	*Simulated Pulse Descriptor Word*	6
SRR	*System Requirements Review*	1, 2, 4
TMAP	*Threat Modeling and Analysis Program*	1
TPOC	*Technical Point of Contact*	4
UAM	*User and Analyst Manual*	6
UML	*Unified Modeling Language*	4

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